SDG7: Ensure access to affordable, reliable, sustainable and modern energy for all

SUMMARY

<u>Original search string: 2,326,063 document results</u>

TITLE-ABS-KEY((energ* OR *power OR renewable* OR {2000 Watt society} OR {smart micro-grid} OR {smart grid} OR {smart microgrid} OR {smart micro grid} OR {smart micro-grids} OR {smart grids} OR {smart microgrids} OR {smart micro grids} OR {smart meter} OR {smart meters} OR electric* OR *fuel* OR {waste conversion} OR Energiewende OR LCA OR {life-cycle assessment} OR {life cycle assessment} OR {life-cycle assessments} OR {life cycle assessments} OR photochemistry OR photovoltaic OR photocatalysis OR {hydrogen production} OR {water splitting} OR {lithium-ion batteries} OR {lithium-ion battery} OR {heat network} OR {district heat} OR {district heating})) AND PUBYEAR BEF 2018 AND PUBYEAR AFT 2012

Graphical summary of co-occurrence of terms from 2000 randomly selected results:

A graphical summary is not available as the SDG tool cannot handle the large number of document results retrieved by the original search query.

<u>Second update search: 317,602 document results</u>

TITLE-ABS-KEY (({energy efficiency} OR {energy consumption} OR {energy transition} OR {clean energy technology} OR {energy equity} OR {energy justice} OR {energy poverty} OR {energy policy} OR renewable* OR {2000 Watt society} OR {smart micro-grid} OR {smart grid} OR {smart microgrid} OR {smart micro-grids} OR {smart grids} OR {smart microgrids} OR {smart meter} OR {smart meters} OR {affordable electricity} OR {electricity consumption} OR {reliable electricity} OR {clean fuel} OR {clean cooking fuel} OR {fuel poverty} OR energiewende OR {life-cycle assessment} OR {life cycle assessment} OR {life-cycle assessments} OR {life cycle assessments} OR ({photochemistry} AND {renewable energy}) OR photovoltaic OR {photocatalytic water splitting} OR {hydrogen production} OR {water splitting} OR {lithium-ion batteries} OR {lithium-ion battery} OR {heat network} OR {district heat} OR {district heating} OR {residential energy consumption} OR {domestic energy consumption} OR {energy security} OR {rural electrification} OR {energy ladder} OR {energy access} OR {energy conservation} OR {low-carbon society} OR {hybrid renewable energy system} OR {hybrid renewable energy systems} OR {fuel switching} OR ({foreign development aid} AND {renewable energy}) OR {energy governance} OR ({official development assistance} AND {electricity}) OR ({energy development} AND {developing countries})) AND NOT ({wireless sensor network} OR {wireless sensor networks})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

PROCESS

The first stage of this review was to compare the original search string to the targets listed under the Sustainable Development Goal (SDG).

[SDG targets listed here: https://sustainabledevelopment.un.org/sdg7](https://sustainabledevelopment.un.org/sdg7)

The purpose of this stage was to understand whether the Scopus results correctly reflected the theme of the SDG. The SDGs were unanimously agreed by member states of the United Nations, with very deliberate language and scope/ exclusions, so the results of this analysis should accurately reflect the specific SDG themes.

SDG 7 aims to ensure **universal access to affordable, reliable and modern energy** such as electricity and an increasing use of clean fuels and technology. **Renewable energy and increased energy efficiency** should be promoted. International financial aid should support the **adoption, research and development of renewable energy** sources, technology and efficiency. Developing countries should be financially supported through **foreign direct investments to adopt and upgrade technology for supplying renewable energy services** and required infrastructure.

A Scopus search was conducted using the original search string. This returned 2,326,063 document results.

The top 20 cited articles were reviewed. 12 of the 20 articles do not closely reflect the theme of the SDG. Specifically, these were related to IT, physics, healthcare, genetics and (nano)engineering. Hits were found on keywords such as energy and power. The eight articles that are closely related to the theme of the SDG primarily cover topics related to advancements in the energy efficiency of solar cells, one covers rechargeable batteries.

Top 20 most cited articles listed in Scopus from the original search string:

*red text shows articles which do not closely match the theme of the SDG

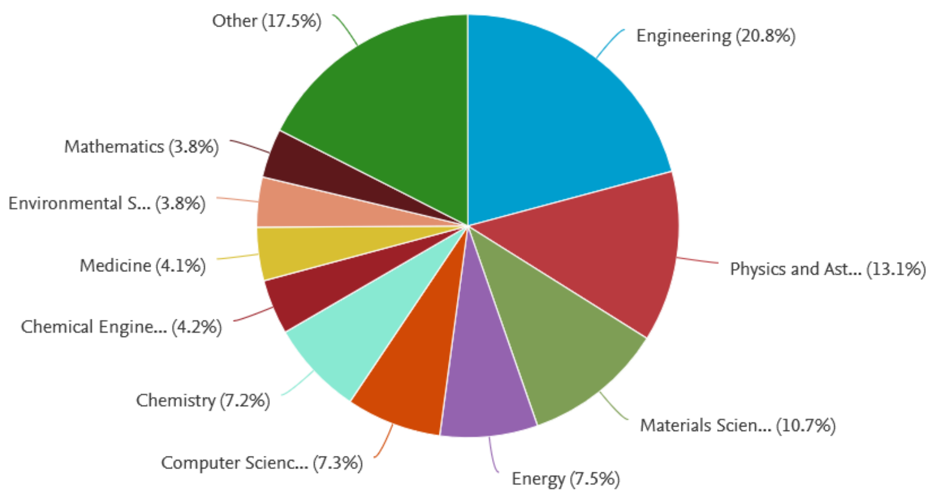
Title	Year	Source title	Cited by	EID
Solar Engineering of Thermal Processes: Fourth Edition	2013	Solar Engineering of Thermal Processes: Fourth Edition	7131	2-s2.0-84891584184
Review of particle physics	2014	Chinese Physics C	5583	2-s2.0-85014877158
Sequential deposition as a route to high-performance perovskite-sensitized solar cells	2013	Nature	5456	2-s2.0-84880507645
Nutritional Epidemiology	2013	Nutritional Epidemiology	4656	2-s2.0-84921763923
Caffe: Convolutional architecture for fast feature embedding	2014	MM 2014 - Proceedings of the 2014 ACM Conference on Multimedia	4629	2-s2.0-84913580146
Efficient planar heterojunction perovskite solar cells by vapour deposition	2013	Nature	4523	2-s2.0-84884411335
The chemistry of two-dimensional layered transition metal dichalcogenide nanosheets	2013	Nature Chemistry	4094	2-s2.0-84875413255
Internet of Things (IoT): A vision, architectural elements, and future directions	2013	Future Generation Computer Systems	3989	2-s2.0-84876943063
Limma powers differential expression analyses for RNA-sequencing and microarray studies	2015	Nucleic Acids Research	3875	2-s2.0-84926507971
Van der Waals heterostructures	2013	Nature	3840	2-s2.0-84881167566
Interface engineering of highly efficient perovskite solar cells	2014	Science	3765	2-s2.0-84905902495

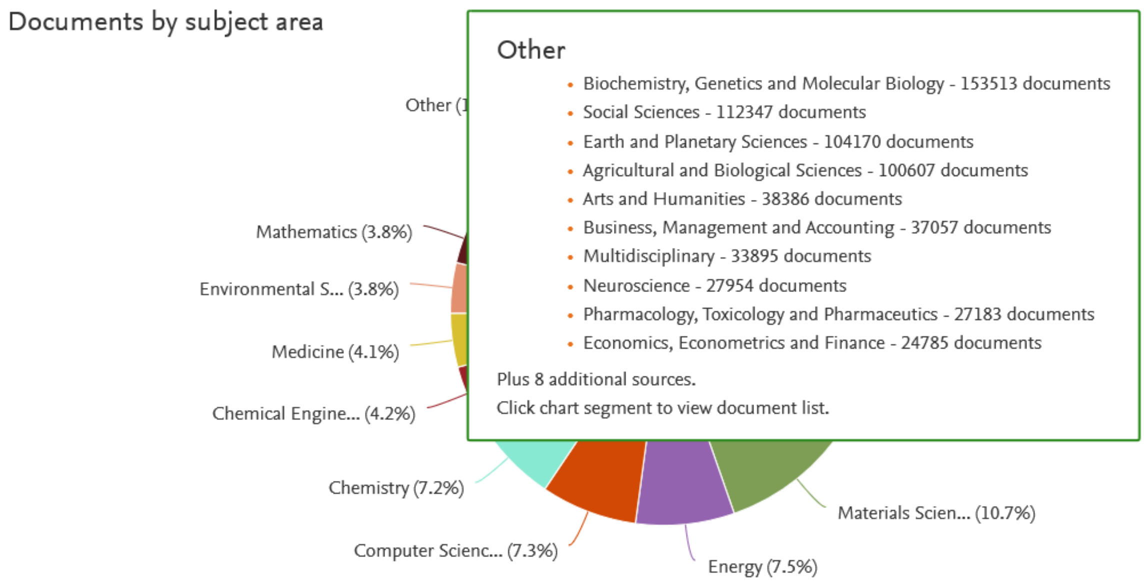
Heart Disease and Stroke Statistics - 2014 Update: A report from the American Heart Association	2014	Circulation	3706	2-s2.0-84893651437
Review of particle physics	2016	Chinese Physics C	3427	2-s2.0-85049178969
Black phosphorus field-effect transistors	2014	Nature Nanotechnology	3424	2-s2.0-84901193930
What will 5G be?	2014	IEEE Journal on Selected Areas in Communications	3410	2-s2.0-84904979683
High-performance photovoltaic perovskite layers fabricated through intramolecular exchange	2015	Science	3400	2-s2.0-84931272007
Planck 2015 results: XIII. Cosmological parameters	2016	Astronomy and Astrophysics	3250	2-s2.0-85020144405
Solvent engineering for high-performance inorganic-organic hybrid perovskite solar cells	2014	Nature Materials	3177	2-s2.0-84928951451
GROMACS 4.5: A high-throughput and highly parallel open source molecular simulation toolkit	2013	Bioinformatics	3171	2-s2.0-84875592758
The Li-ion rechargeable battery: A perspective	2013	Journal of the American Chemical Society	3058	2-s2.0-84873825623

A randomised sample of 20 articles could not be analysed as the total number of document results is too high to be able to use the SDG tool.

Reflection on subject areas that should be excluded (incl. pie charts)

Documents by subject area





The analyses show that articles are retrieved from a variety of fields. Whereas this is not a problem per se – as the SDG is covered by multiple fields such as engineering, energy and social sciences – the analyses also show that articles and fields are included that do not relate to the theme of the SDG. The general analysis of the overall document results shows that the field of ‘Engineering’ delivers about 20% of the articles retrieved. This could be explained by articles on energy efficiency and advances in engineering techniques. The field of ‘Physics and astronomy’ is the second largest field (disregarding the ‘other’ field), but not relevant for this SDG. A variety of other fields contribute to the document results and several of them are not directly relevant, e.g. ‘Computer Sciences’, ‘Medicine’ and the subfields collated under ‘Other’ such as ‘Biochemistry, Genetics and Molecular Biology’. ‘Energy’ forms a smaller contribution, about 7,5%, but it could be expected that its contribution should be strengthened as energy is the main theme of this SDG. Furthermore, it could be expected that more document results from the field of ‘Social sciences’ should be retrieved due to targets on energy equity and financial aid. Therefore, improvements are needed, and a first step would be to narrow down keywords to retrieve articles in fields that are closely related to the theme of the SDG. A next step would be to add keywords on articles related to targets that are currently missing in the document results.

Update 1) The search keywords were edited to remove the broader search terms, or to make them more specific by combining with other search keywords.

<u>Original search string: 2,326,063 document results</u>

TITLE-ABS-KEY((energ* OR *power OR renewable* OR {2000 Watt society} OR {smart micro-grid} OR {smart grid} OR {smart microgrid} OR {smart micro grid} OR {smart micro-grids} OR {smart grids} OR {smart microgrids} OR {smart micro grids} OR {smart meter} OR {smart meters} OR electric* OR *fuel* OR {waste conversion} OR Energiewende OR LCA OR {life-cycle assessment} OR {life cycle assessment} OR {life-cycle assessments} OR {life cycle assessments} OR photochemistry OR photovoltaic OR photocatalysis OR {hydrogen production} OR {water splitting} OR {lithium-ion batteries} OR {lithium-ion battery} OR {heat network} OR {district heat} OR {district heating})) AND PUBYEAR BEF 2018 AND PUBYEAR AFT 2012

Rationale for edits to the search keywords:

<u>query term</u>	**<u>decision</u>**	**<u>rationale</u>**
Energ*	Changed to {energy efficiency} {energy consumption} {energy	Energ* is too broad and led to hits on articles in physics, healthcare, IT and biology. Narrowed down to keywords related to the SDG.

	transition} {clean energy technology} {energy equity} {energy justice} {energy poverty} {energy policy}	
*power	Deleted	Not directly related to the theme of the SDG and power in terms of energy is already covered by other keywords. This keyword also led to hits in the field of IT. Deleted as it does not seem to add value to the query at this moment.
{smart micro grid}	Deleted	Deleted to clean up the query, variants are still included
{smart micro grids}	Deleted	Deleted to clean up the query, variants are still included
electric*	Changed to {affordable electricity} {electricity consumption} {reliable electricity}	Narrowed down as the keyword is too broad and led to hits in a variety of fields
fuel	Changed to {clean fuel} {clean cooking fuel} {fuel poverty}	*fuel* is too broad, narrowed down to keywords that relate to targets of the SDG
{waste conversion}	Deleted	Will lead to hits on waste instead of electricity / energy
LCA	Deleted	Results should come up when using the other variants, deleted to clean up the query
Photochemistry	Changed to ({photochemistry} AND {renewable energy})	Narrowed down as photochemistry is also a common keyword in the field of biology or other fields of chemistry other than chemical engineering in the field of renewable energies
Photocatalysis	Changed to {photocatalytic water splitting}	Narrowed down to {photocatalytic water splitting} as photocatalysis can involve a variety of processes that do not relate to renewable energy, such metal or water treatment
AND NOT {wireless sensor networks}	Added	Added as {energy efficiency} is also often mentioned in relation to IT and WSNs. {energy efficiency} is a core target of this SDG so this cannot be deleted, added this query term as a first step to exclude a majority of IT articles

Updated search string (first update): 300,098 document results

TITLE-ABS-KEY (({energy efficiency} OR {energy consumption} OR {energy transition} OR {clean energy technology} OR {energy equity} OR {energy justice} OR {energy poverty} OR {energy policy} OR renewable* OR {2000 Watt society} OR {smart micro-grid} OR {smart grid} OR {smart microgrid} OR {smart micro-grids} OR {smart grids} OR {smart microgrids} OR {smart meter} OR {smart meters} OR {affordable electricity} OR {electricity consumption} OR {reliable electricity} OR {clean fuel} OR {clean cooking fuel} OR {fuel poverty} OR energiewende OR {life-cycle assessment} OR {life cycle assessment} OR {life-cycle assessments} OR {life cycle assessments} OR ({photochemistry} AND {renewable energy}) OR photovoltaic OR {photocatalytic water splitting} OR {hydrogen production} OR {water splitting} OR {lithium-ion batteries} OR {lithium-ion battery} OR {heat network} OR {district heat} OR {district heating}) AND NOT ({wireless sensor network} OR {wireless sensor networks})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

The results of the updated search were reviewed to check whether they aligned closely with the SDG theme.

The updated search returned 300,098 document results, compared to 2,326,063 document results for the original search.

The top 20 cited articles were reviewed. Two articles do not closely relate to the theme of the SDG. These two articles cover topics related to IT and energy efficiency. The other 18 articles mainly cover topics related to the engineering aspects of renewable energy (solar cells, water splitting) or batteries.

Top 20 most cited articles listed in Scopus from search string version 1:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Solar Engineering of Thermal Processes: Fourth Edition	2013	Solar Engineering of Thermal Processes: Fourth Edition	7131	2-s2.0-84891584184
Sequential deposition as a route to high-performance perovskite-sensitized solar cells	2013	Nature	5456	2-s2.0-84880507645
Efficient planar heterojunction perovskite solar cells by vapour deposition	2013	Nature	4523	2-s2.0-84884411335
What will 5G be?	2014	IEEE Journal on Selected Areas in Communications	3410	2-s2.0-84904979683
High-performance photovoltaic perovskite layers fabricated through intramolecular exchange	2015	Science	3400	2-s2.0-84931272007
Solvent engineering for high-performance inorganic-organic hybrid perovskite solar cells	2014	Nature Materials	3177	2-s2.0-84928951451
The Li-ion rechargeable battery: A perspective	2013	Journal of the American Chemical Society	3058	2-s2.0-84873825623
Compositional engineering of perovskite materials for high-performance solar cells	2015	Nature	2873	2-s2.0-84922586427
Massive MIMO for next generation wireless systems	2014	IEEE Communications Magazine	2713	2-s2.0-84896823919
A polymer tandem solar cell with 10.6% power conversion efficiency	2013	Nature Communications	2351	2-s2.0-84874611797
Sodium-ion batteries	2013	Advanced Functional Materials	2263	2-s2.0-84873405642
Advanced Engineering Thermodynamics	2016	Advanced Engineering Thermodynamics	2152	2-s2.0-85019442444
Iodide management in formamidinium-lead-halide-based perovskite layers for efficient solar cells	2017	Science	2106	2-s2.0-85021664119
Cesium-containing triple cation perovskite solar cells: Improved stability, reproducibility and high efficiency	2016	Energy and Environmental Science	1938	2-s2.0-84974593527
Where do batteries end and supercapacitors begin?	2014	Science	1923	2-s2.0-84903362570
Towards greener and more sustainable batteries for electrical energy storage	2015	Nature Chemistry	1810	2-s2.0-84924528297
Electron-hole diffusion lengths > 175 μm in solution-grown CH3NH3PbI3 single crystals	2015	Science	1806	2-s2.0-84923886273

Metal-free efficient photocatalyst for stable visible water splitting via a two-electron pathway	2015	Science	1716	2-s2.0-84923862249
High-efficiency solution-processed perovskite solar cells with millimeter-scale grains	2015	Science	1707	2-s2.0-84921890919
Perovskite solar cells with a planar heterojunction structure prepared using room-temperature solution processing techniques	2014	Nature Photonics	1686	2-s2.0-84893299607

In addition, a randomised sample of 20 articles was reviewed. Seven of the 20 articles were not closely related to the theme of the SDG. These seven articles cover topics in the field of chemistry, hydrology, engineering and IT. Hits were found on the keywords ‘energy efficiency’, ‘energy consumption’ and ‘renewable energy’.

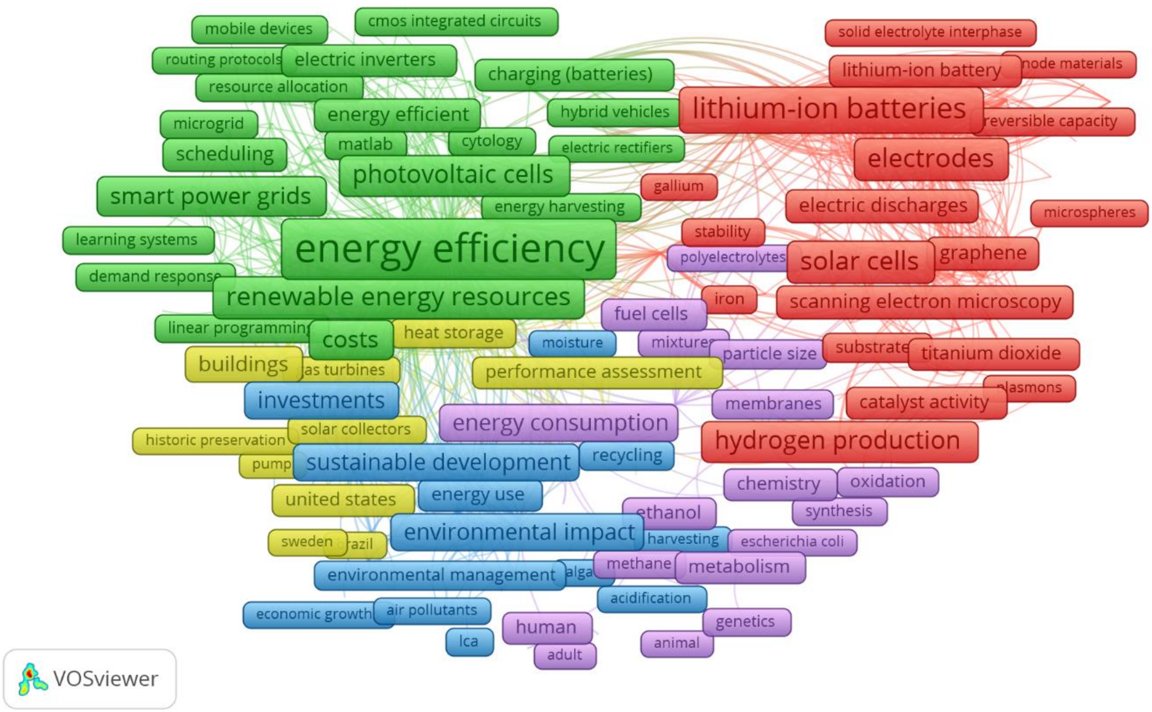
Randomised sample of 20 articles listed in Scopus from search string version 1:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Long-Cycling aqueous organic Redox flow battery (AORFB) toward sustainable and safe energy storage	2017	Journal of the American Chemical Society	115	2-s2.0-85019070090
MoS2/C nanosheets Encapsulated Sn@SnOx nanoparticles as high-performance Lithium-iom battery anode material	2016	Electrochimica Acta	29	2-s2.0-84962484834
A yellow emitting phosphor Dy:Bi4Si3O12 crystal for LED application	2014	Materials Letters	22	2-s2.0-84906549808
Rapid hydrothermal and post-calcination synthesis of well-shaped LiNi0.5Mn1.5O4cathode materials for lithium ion batteries	2017	Journal of Alloys and Compounds	17	2-s2.0-85014090880
The Limit Deposit Velocity model, a new approach	2015	Journal of Hydrology and Hydromechanics	6	2-s2.0-84945962740
A compact internal drum test rig for measurements of rolling contact forces between a single tread block and a substrate	2017	Measurement: Journal of the International Measurement Confederation	4	2-s2.0-84994633128
Feasibility study of biodiesel production from residual oil of palm oil mill effluent	2017	International Journal of GEOMATE	3	2-s2.0-85018299884
Improve electrochemical performance of LiFePO4/C cathode by coating Ti2O3 through a facile route	2015	Ionics	3	2-s2.0-84947613353
A user-centric space heating energy management framework for multi-family residential facilities based on occupant pattern prediction modeling	2017	Building Simulation	2	2-s2.0-85030869314
The accuracy of wind energy forecasts and prospects for improvement	2017	International Conference on the European Energy Market, EEM	1	2-s2.0-85027133305
Battery storage application for covering the mismatch between scheduled and CHP plant production	2017	2017 52nd International Universities Power Engineering	0	2-s2.0-85046069180

		Conference, UPEC 2017		
Streaming sensor data from dynamically reprogrammable tasks running on mobile devices	2017	BuildSys 2017 - Proceedings of the 4th ACM International Conference on Systems for Energy-Efficient Built Environments	0	2-s2.0-85052012675
A dynamic voltage-combiners biased OTA for low-power and high-speed SC circuits	2017	PRIME 2017 - 13th Conference on PhD Research in Microelectronics and Electronics, Proceedings	0	2-s2.0-85027579238
A system for multipoint measurement of wind parameters for optimizing the location of small wind turbines	2017	International Multidisciplinary Scientific GeoConference Surveying Geology and Mining Ecology Management, SGEM	0	2-s2.0-85063076955
Solid-liquid pipeline flow - from menagerie to mechanistic modelling.	2017		0	2-s2.0-85040265055
Trap states comparison of mesoporous TiO_2 photoanodes with different particle sizes	2016	Journal of Nanoscience and Nanotechnology	0	2-s2.0-84974738173
The use of algorithms for light control	2016	Start-Up Creation: The Smart Eco-Efficient Built Environment	0	2-s2.0-85011791522
Decomposition of fluctuating photovoltaic generation power in frequency bands and analysis of chaotic properties	2016	IEEJ Transactions on Power and Energy	0	2-s2.0-84977139244
Installation of shore approaches and sea-lines using trenchless methods: Technologies and case studies	2015	Recent Progress in Desalination, Environmental and Marine Outfall Systems	0	2-s2.0-84955421356
Retro-commissioning to improve energy efficiency in high-performing buildings	2015	Energy Engineering: Journal of the Association of Energy Engineering	0	2-s2.0-84929581588

Graphical summary of co-occurrence of terms from 2000 randomly selected results:



The new search results were compared to the original search results to check if the updated search was primarily removing irrelevant results.

Combining the first updated search query with the original search using AND NOT function in Scopus resulted in 2,025,965 document results – i.e. the results that were excluded from the original search in the new query. Eighteen of the 20 articles do not closely reflect the theme of the SDG. Two articles were excluded which are relevant for this SDG, these cover energy efficiency and engineering regarding solar cells.

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Review of particle physics	2014	Chinese Physics C	5583	2-s2.0-85014877158
Nutritional Epidemiology	2013	Nutritional Epidemiology	4656	2-s2.0-84921763923
Caffe: Convolutional architecture for fast feature embedding	2014	MM 2014 - Proceedings of the 2014 ACM Conference on Multimedia	4629	2-s2.0-84913580146
The chemistry of two-dimensional layered transition metal dichalcogenide nanosheets	2013	Nature Chemistry	4094	2-s2.0-84875413255
Internet of Things (IoT): A vision, architectural elements, and future directions	2013	Future Generation Computer Systems	3989	2-s2.0-84876943063
Limma powers differential expression analyses for RNA-sequencing and microarray studies	2015	Nucleic Acids Research	3875	2-s2.0-84926507971

Van der Waals heterostructures	2013	Nature	3840	2-s2.0-84881167566
Interface engineering of highly efficient perovskite solar cells	2014	Science	3765	2-s2.0-84905902495
Heart Disease and Stroke Statistics - 2014 Update: A report from the American Heart Association	2014	Circulation	3706	2-s2.0-84893651437
Review of particle physics	2016	Chinese Physics C	3427	2-s2.0-85049178969
Black phosphorus field-effect transistors	2014	Nature Nanotechnology	3424	2-s2.0-84901193930
Planck 2015 results: XIII. Cosmological parameters	2016	Astronomy and Astrophysics	3250	2-s2.0-85020144405
GROMACS 4.5: A high-throughput and highly parallel open source molecular simulation toolkit	2013	Bioinformatics	3171	2-s2.0-84875592758
Visible light photoredox catalysis with transition metal complexes: Applications in organic synthesis	2013	Chemical Reviews	2997	2-s2.0-84880124916
Standards of medical care in diabetes-2014	2014	Diabetes Care	2893	2-s2.0-84892649479
Heart disease and stroke statistics-2016 update a report from the American Heart Association	2016	Circulation	2862	2-s2.0-84950104119
Phosphorene: An unexplored 2D semiconductor with a high hole mobility	2014	ACS Nano	2729	2-s2.0-84898060562
Dye-sensitized solar cells with 13% efficiency achieved through the molecular engineering of porphyrin sensitizers	2014	Nature Chemistry	2729	2-s2.0-84895119853
Raman spectroscopy as a versatile tool for studying the properties of graphene	2013	Nature Nanotechnology	2662	2-s2.0-84876373110
Carbon nanotubes: Present and future commercial applications	2013	Science	2606	2-s2.0-84873808704

The analyses show that the first update successfully excludes a majority of non-relevant fields, especially regarding healthcare and physics. Articles in the field of IT, chemistry and engineering which do not relate to the SDG are still (partly) included due to hits on keywords such as 'energy efficiency' and 'energy consumption'. The latter keyword could be narrowed down to a keyword that retrieves articles that match the targets of the SDG. The analyses also show several targets are still missing in the search query and the documents retrieved by the query, especially regarding financial aid, development and universal access. The current query focusses on the energy efficiency and R&D targets of the SDG. A next update should aim to add keywords on missing or incomplete targets.

Update 2) A second update was made to the search string to cover missing or incomplete targets.

To bring these articles into the search results, the following were added to the search string:

Target	Keywords to be added / narrowed down
By 2030, ensure **universal access to affordable, reliable and modern energy services**	{energy consumption} changed to {residential energy consumption} OR {domestic energy consumption} {energy security} {rural electrification} {energy ladder} {energy access}

By 2030, **increase** substantially the share of **renewable energy** in the global energy mix	{fuel switching} {low-carbon society} {hybrid renewable energy system} {hybrid renewable energy systems}
By 2030, double the global rate of improvement in **energy efficiency**	{energy conservation}
By 2030, enhance international cooperation to **facilitate access to clean energy research and technology**, including renewable energy, energy efficiency and advanced and cleaner fossil-fuel technology, and **promote investment in energy infrastructure and clean energy technology**	Included
By 2030, **expand infrastructure and upgrade technology for supplying modern and sustainable energy services for all in developing countries**, in particular least developed countries, small island developing States, and land-locked developing countries, in accordance with their respective programmes of support	({foreign development aid} AND {renewable energy}) {energy governance} ({official development assistance} AND {electricity}) ({energy development} AND {developing countries})

Second update search: 317,602 document results

TITLE-ABS-KEY (({energy efficiency} OR {energy consumption} OR {energy transition} OR {clean energy technology} OR {energy equity} OR {energy justice} OR {energy poverty} OR {energy policy} OR renewable* OR {2000 Watt society} OR {smart micro-grid} OR {smart grid} OR {smart microgrid} OR {smart micro-grids} OR {smart grids} OR {smart microgrids} OR {smart meter} OR {smart meters} OR {affordable electricity} OR {electricity consumption} OR {reliable electricity} OR {clean fuel} OR {clean cooking fuel} OR {fuel poverty} OR energiewende OR {life-cycle assessment} OR {life cycle assessment} OR {life-cycle assessments} OR {life cycle assessments} OR ({photochemistry} AND {renewable energy}) OR photovoltaic OR {photocatalytic water splitting} OR {hydrogen production} OR {water splitting} OR {lithium-ion batteries} OR {lithium-ion battery} OR {heat network} OR {district heat} OR {district heating} OR {residential energy consumption} OR {domestic energy consumption} OR {energy security} OR {rural electrification} OR {energy ladder} OR {energy access} OR {energy conservation} OR {low-carbon society} OR {hybrid renewable energy system} OR {hybrid renewable energy systems} OR {fuel switching} OR ({foreign development aid} AND {renewable energy}) OR {energy governance} OR ({official development assistance} AND {electricity}) OR ({energy development} AND {developing countries})) AND NOT ({wireless sensor network} OR {wireless sensor networks})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

The results of the updated search were reviewed to check whether they aligned closely with the SDG theme.

The top 20 cited articles were reviewed, this is the same top 20 as retrieved with search query v1. Two of the 20 articles are not closely related to the theme of the SDG. Both articles cover topics in the field of IT and were retrieved through hits on the keyword ‘energy efficiency’.

Top 20 most cited articles listed in Scopus from search string version 2:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Solar Engineering of Thermal Processes: Fourth Edition	2013	Solar Engineering of Thermal	7131	2-s2.0-84891584184

		Processes: Fourth Edition		
Sequential deposition as a route to high-performance perovskite-sensitized solar cells	2013	Nature	5456	2-s2.0-84880507645
Efficient planar heterojunction perovskite solar cells by vapour deposition	2013	Nature	4523	2-s2.0-84884411335
What will 5G be?	2014	IEEE Journal on Selected Areas in Communications	3410	2-s2.0-84904979683
High-performance photovoltaic perovskite layers fabricated through intramolecular exchange	2015	Science	3400	2-s2.0-84931272007
Solvent engineering for high-performance inorganic-organic hybrid perovskite solar cells	2014	Nature Materials	3177	2-s2.0-84928951451
The Li-ion rechargeable battery: A perspective	2013	Journal of the American Chemical Society	3058	2-s2.0-84873825623
Compositional engineering of perovskite materials for high-performance solar cells	2015	Nature	2873	2-s2.0-84922586427
Massive MIMO for next generation wireless systems	2014	IEEE Communications Magazine	2713	2-s2.0-84896823919
A polymer tandem solar cell with 10.6% power conversion efficiency	2013	Nature Communications	2351	2-s2.0-84874611797
Sodium-ion batteries	2013	Advanced Functional Materials	2263	2-s2.0-84873405642
Advanced Engineering Thermodynamics	2016	Advanced Engineering Thermodynamics	2152	2-s2.0-85019442444
Iodide management in formamidinium-lead-halide-based perovskite layers for efficient solar cells	2017	Science	2106	2-s2.0-85021664119
Cesium-containing triple cation perovskite solar cells: Improved stability, reproducibility and high efficiency	2016	Energy and Environmental Science	1938	2-s2.0-84974593527
Where do batteries end and supercapacitors begin?	2014	Science	1923	2-s2.0-84903362570
Towards greener and more sustainable batteries for electrical energy storage	2015	Nature Chemistry	1810	2-s2.0-84924528297
Electron-hole diffusion lengths > 175 μm in solution-grown CH3NH3PbI3 single crystals	2015	Science	1806	2-s2.0-84923886273
Metal-free efficient photocatalyst for stable visible water splitting via a two-electron pathway	2015	Science	1716	2-s2.0-84923862249
High-efficiency solution-processed perovskite solar cells with millimeter-scale grains	2015	Science	1707	2-s2.0-84921890919
Perovskite solar cells with a planar heterojunction structure prepared using room-temperature solution processing techniques	2014	Nature Photonics	1686	2-s2.0-84893299607

In addition, a randomised sample of 20 articles was reviewed. Four of the 20 articles do not closely reflect the theme of the SDG, all of these relate to engineering. These were retrieved through hits on 'energy efficiency' and 'renewable*.

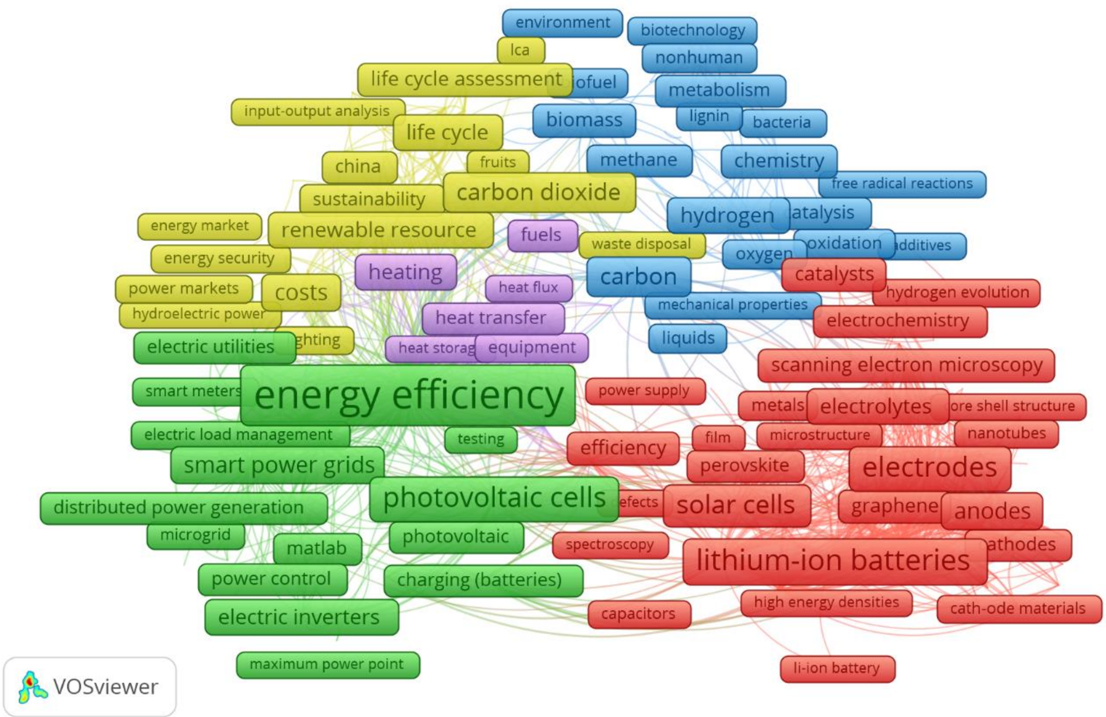
Randomised sample of 20 articles listed in Scopus from search string version 2:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Study on a novel pressurized MCFC hybrid system with CO2 capture	2016	Energy	18	2-s2.0-84979924157
Innovative Approach for Heating of Buildings Using Water from a Flooded Coal Mine Through an Open Loop Based Single Shaft GSHP System	2015	Energy Procedia	11	2-s2.0-84947093889
A robust high order sliding mode for buck-boost converters with large disturbances	2015	Zhongguo Dianji Gongcheng Xuebao/Proceedings of the Chinese Society of Electrical Engineering	10	2-s2.0-84927723388
Spin-coating process of an inorganic Sb2S3 thin film for photovoltaic applications	2016	Journal of Nanoscience and Nanotechnology	8	2-s2.0-84990934000
Active and reactive power flow analysis of a STATCOM with virtual synchronous machine control	2015	2015 IEEE 16th Workshop on Control and Modeling for Power Electronics, COMPEL 2015	6	2-s2.0-84957899775
Energy integration of a hydrotreating process for the production of biojet fuel	2016	Computer Aided Chemical Engineering	5	2-s2.0-84994200742
Long-Term Planning with Battery-Based Energy Storage Transportation in Power System	2017	IEEE Green Technologies Conference	4	2-s2.0-85019746161
Energy efficiency analysis of insulation and warming for biogas engineering in cold area	2015	Taiyangneng Xuebao/Acta Energiae Solaris Sinica	3	2-s2.0-84937410045
Testing for the size heuristic in householders' perceptions of energy consumption	2017	Journal of Environmental Psychology	2	2-s2.0-85032269605
Causal interactions between environmental degradation, renewable energy, nuclear energy and real GDP: a dynamic panel data approach	2017	Environment Systems and Decisions	2	2-s2.0-85000444094
Optical quantum technology in space using small satellites	2017	Proceedings of the International Astronautical Congress, IAC	2	2-s2.0-85051394595
Investigation of bio-based aromatic acids as corrosion inhibitor	2015	NACE - International Corrosion Conference Series	2	2-s2.0-84940544713
Preparation and properties of cotton stalk bark fibers using combined steam explosion and laccase treatment	2017	Journal of Applied Polymer Science	1	2-s2.0-85018961289
Investigation of the thermal performance of green roof on a mild warm climate	2016	International Journal of Renewable Energy Research	1	2-s2.0-84977604183
Performance assessment of a wind driven membrane desalination unit in Saudi Arabia	2017	Journal of Engineering Research	0	2-s2.0-85026641151
Thermal abuse of Lithium-ion batteries: Combined study by experiments and simulation	2017	EVS 2017 - 30th International Electric Vehicle Symposium and Exhibition	0	2-s2.0-85050114896

International Conference on Emerging Trends in Electrical, Electronics and Sustainable Energy Systems, ICETEESES 2016	2016	International Conference on Emerging Trends in Electrical, Electronics and Sustainable Energy Systems, ICETEESES 2016	0	2-s2.0-84995537944
Testing of pyromagTM For energy efficient hot magnetic seperation	2016	IMPC 2016 - 28th International Mineral Processing Congress	0	2-s2.0-85048347294
Efficiency enhancement of organic photovoltaic modules prepared via wash processing	2015	Solar Energy Materials and Solar Cells	0	2-s2.0-84936768045
On-Orbit measurement of next generation space solar cell technology on the international space station	2015	13th International Energy Conversion Engineering Conference	0	2-s2.0-84946761143

Graphical summary of co-occurrence of terms from 2000 randomly selected results:



The analyses and the co-occurrence of the keywords show that there is still an overly focus on energy efficiency and research on renewable energy sources. The keywords added in this update led to a higher number of document results, but the results are not easily identifiable in the analyses. This can be partially explained as advancements made regarding energy efficiency of renewable energy sources is a hot topic. However, future updates and more research on which keywords to use is required to add more articles on targets in relation to universal access to energy services. In addition, the keyword 'renewable*' leads to unwanted hits and should be narrowed down in a future update.